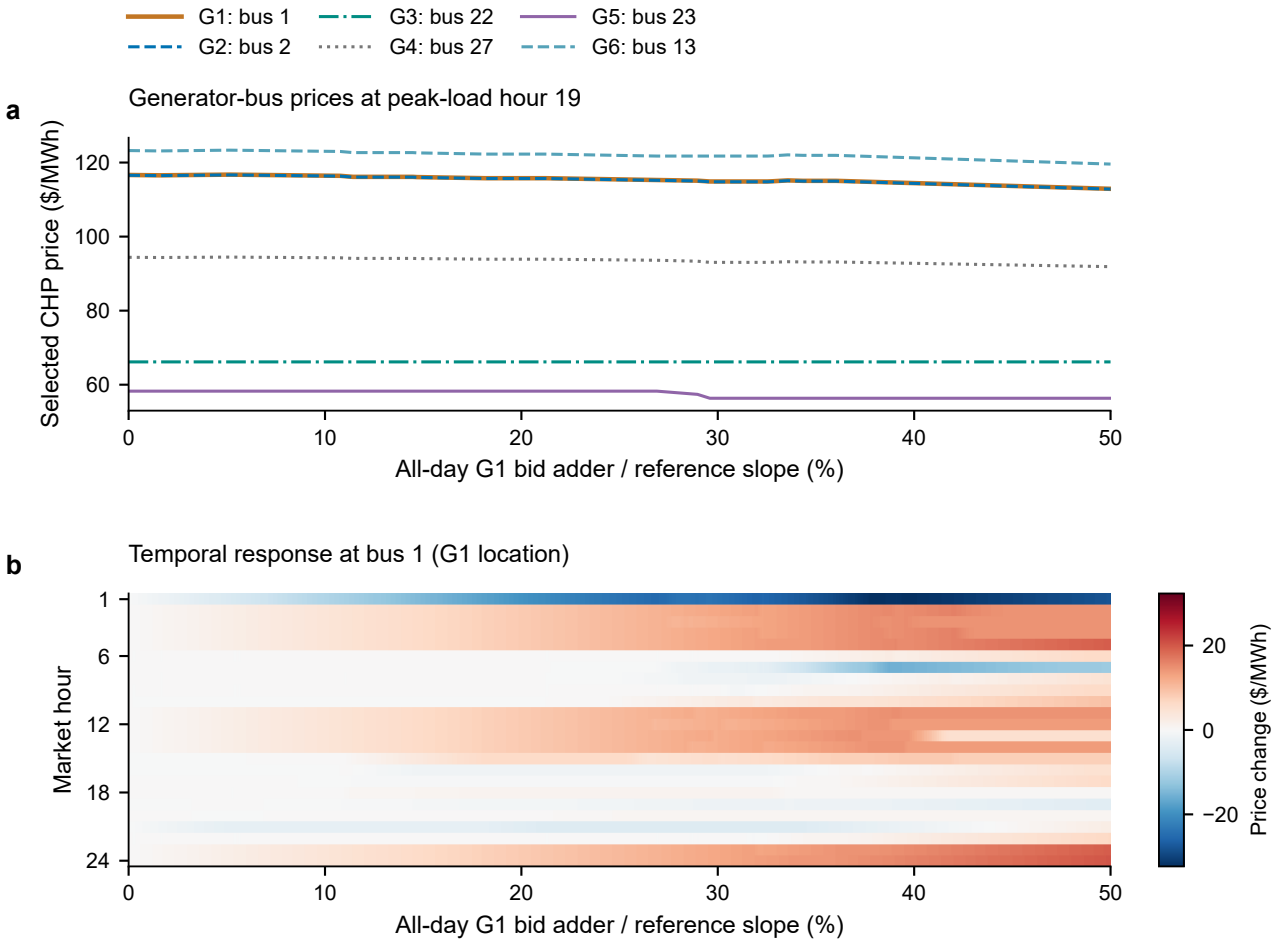
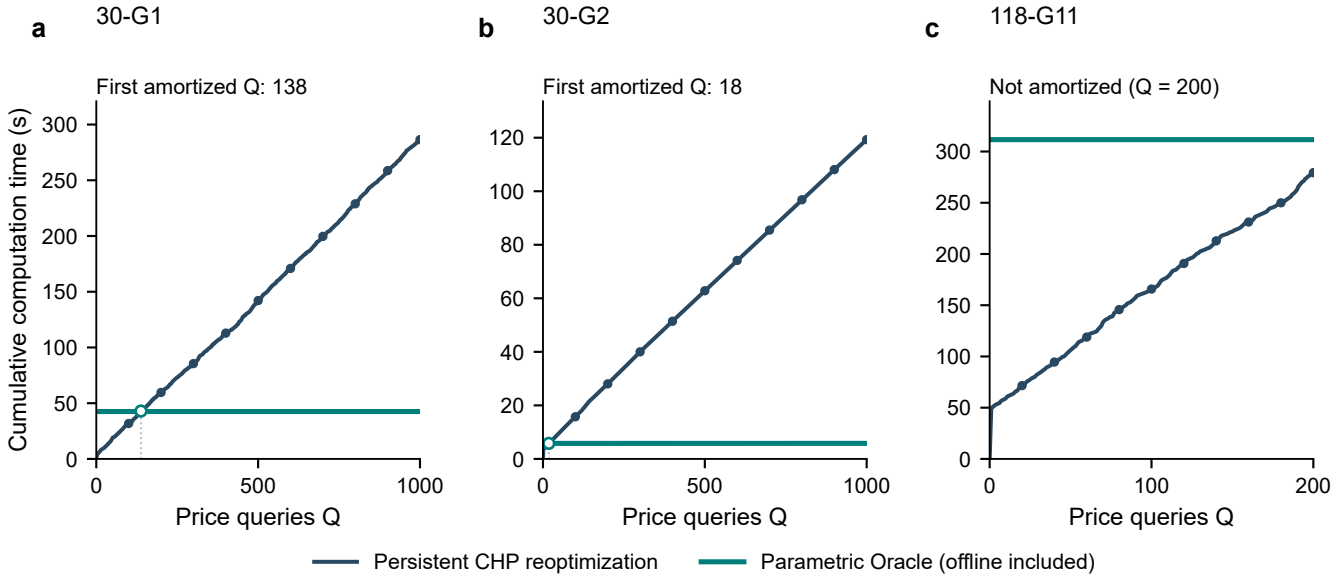
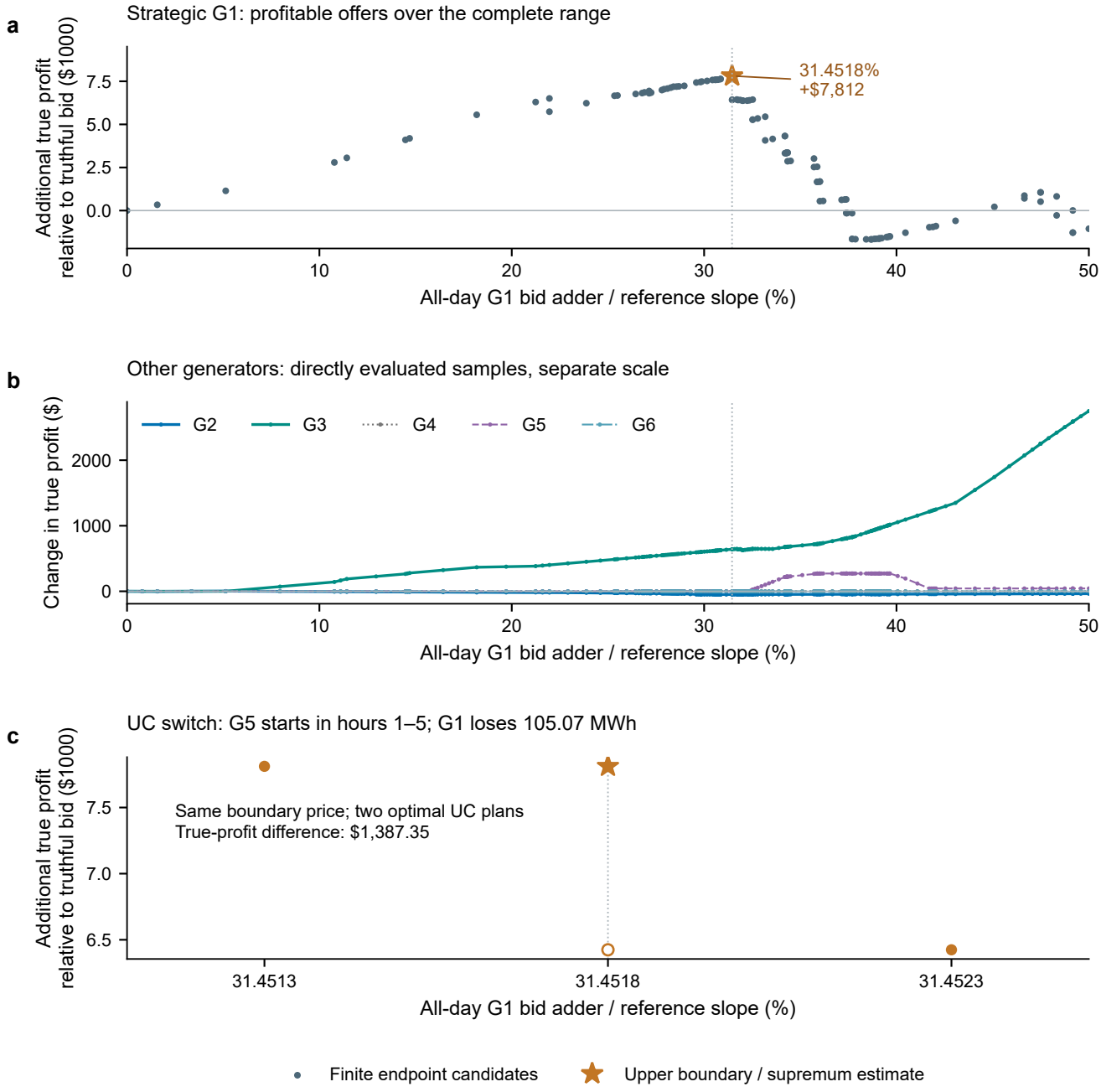
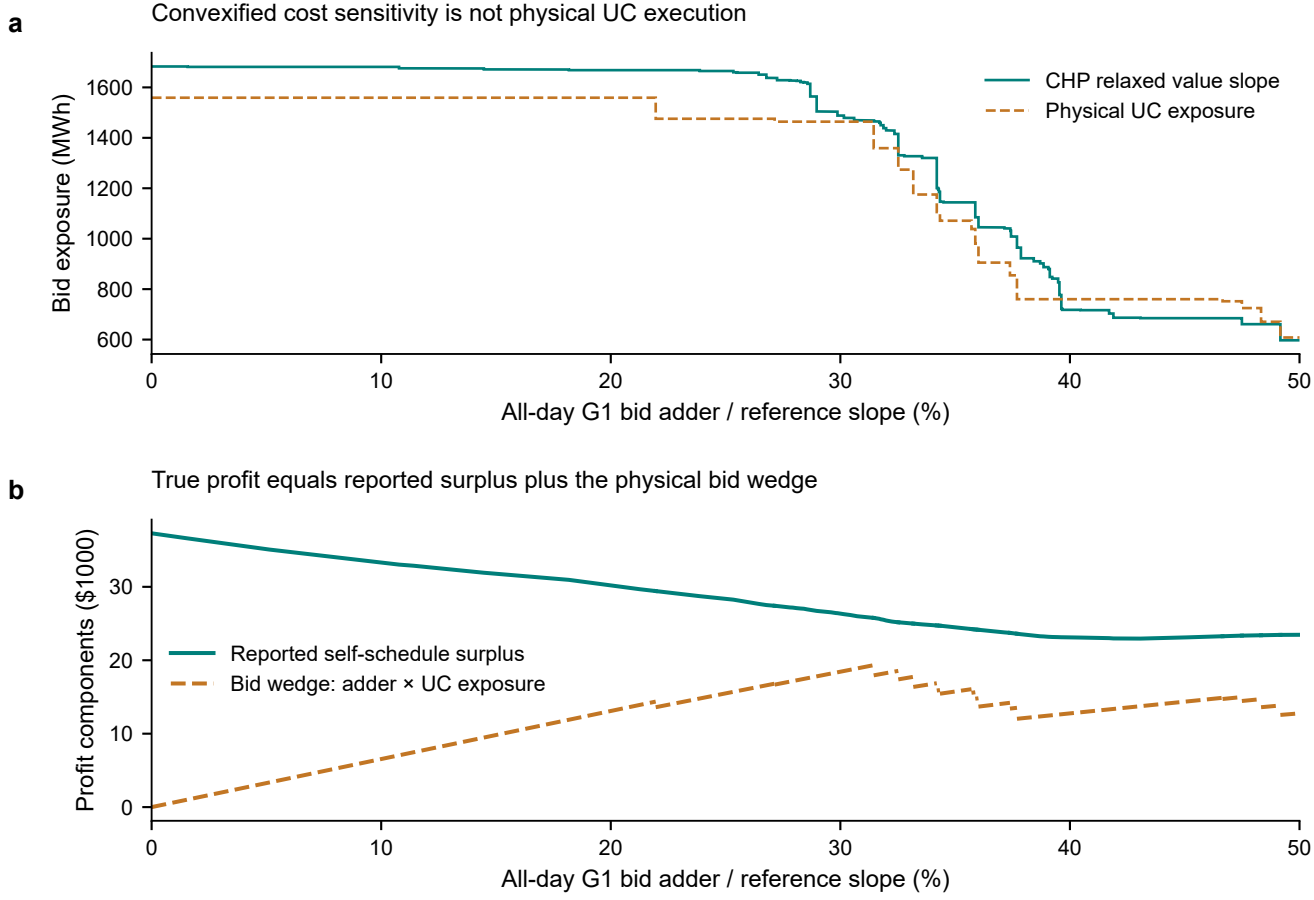
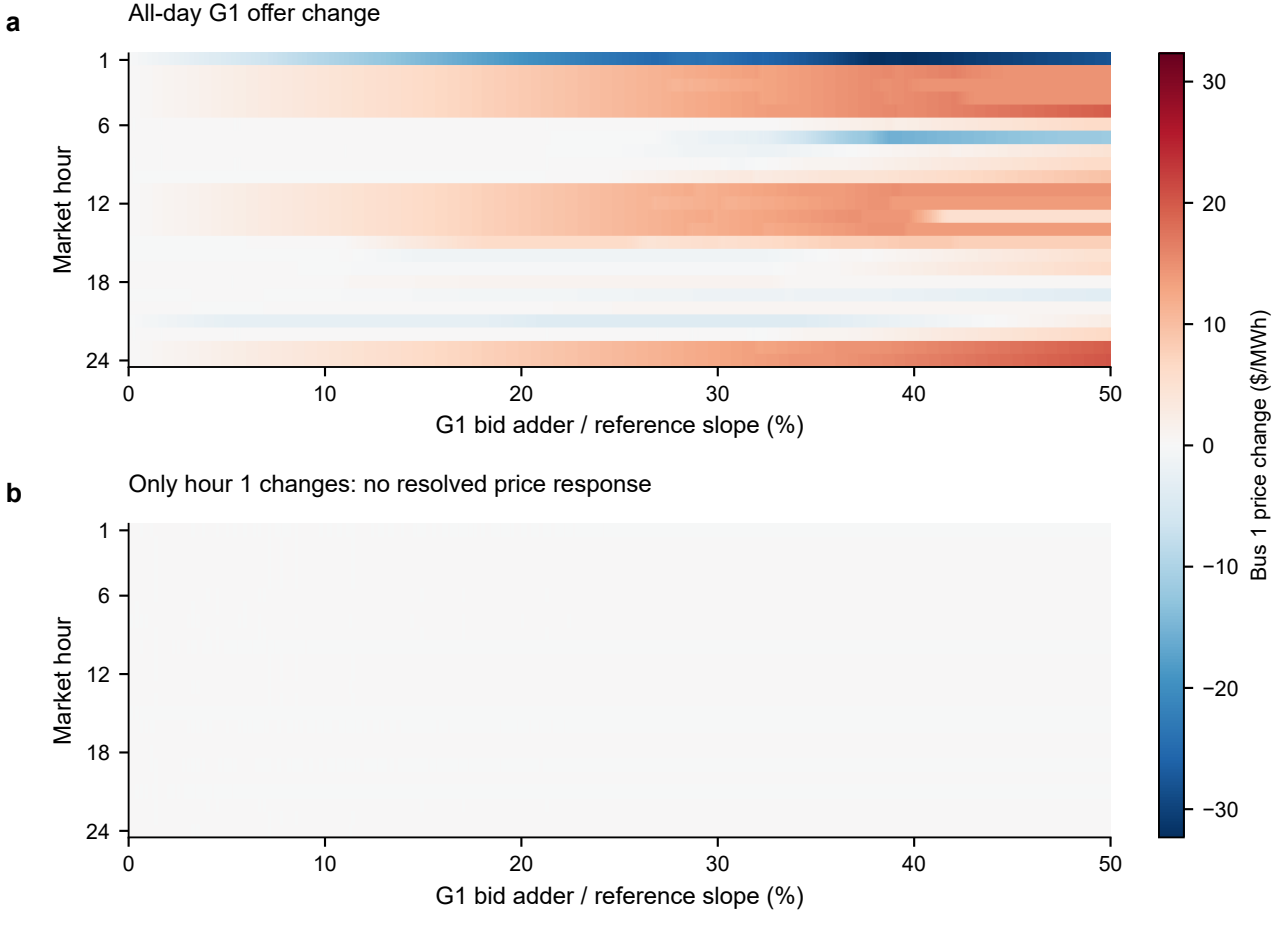

G1: bus 1
G2: bus 2
G3: bus 22
G4: bus 27
G5: bus 23
G6: bus 13
a
Generator-bus prices at peak-load hour 19
Selected CHP price ($/MWh)
120
100
80
60
0
10
20
30
40
50
All-day G1 bid adder / reference slope (%)
b
Temporal response at bus 1 (G1 location)
Market hour
1
6
12
18
24
0
10
20
30
40
50
All-day G1 bid adder / reference slope (%)
20
0
−20
Price change ($/MWh)


a
30-G1
First amortized Q: 138
b
30-G2
First amortized Q: 18
c
118-G11
Not amortized (Q = 200)
Cumulative computation time (s)
Price queries Q
Price queries Q
Price queries Q
Persistent CHP reoptimization
Parametric Oracle (offline included)


a
Strategic G1: profitable offers over the complete range
Additional true profit relative to truthful bid ($1000)
7.5
5.0
2.5
0.0
31.4518%
+$7,812
0
10
20
30
40
50
All-day G1 bid adder / reference slope (%)
b
Other generators: directly evaluated samples, separate scale
Change in true profit ($)
G2
G3
G4
G5
G6
2000
1000
0
0
10
20
30
40
50
All-day G1 bid adder / reference slope (%)
c
UC switch: G5 starts in hours 1–5; G1 loses 105.07 MWh
Additional true profit relative to truthful bid ($1000)
Same boundary price; two optimal UC plans
True-profit difference: $1,387.35
7.5
7.0
6.5
31.4513
31.4518
31.4523
All-day G1 bid adder / reference slope (%)
Finite endpoint candidates
Upper boundary / supremum estimate


a
Convexified cost sensitivity is not physical UC execution
Bid exposure (MWh)
1600
1400
1200
1000
800
600
0
10
20
30
40
50
All-day G1 bid adder / reference slope (%)
CHP relaxed value slope
Physical UC exposure
b
True profit equals reported surplus plus the physical bid wedge
Profit components ($1000)
30
20
10
0
0
10
20
30
40
50
All-day G1 bid adder / reference slope (%)
Reported self-schedule surplus
Bid wedge: adder × UC exposure


a
All-day G1 offer change
Market hour
1
6
12
18
24
0
10
20
30
40
50
G1 bid adder / reference slope (%)
b
Only hour 1 changes: no resolved price response
Market hour
1
6
12
18
24
0
10
20
30
40
50
G1 bid adder / reference slope (%)
30
20
10
0
−10
−20
−30
Bus 1 price change ($/MWh)